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Perceptions of Mobile Green in urban spaces with contrasting visibility and green context

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Temporary urban greening may be especially valuable in dense cities where permanent greening is constrained by limited space, structural conditions, or competing land uses, yet its reception in different urban contexts remains insufficiently understood. This study examines user responses to "Mobile Green," a movable tree-based greening system designed for temporary deployment on artificial ground. Questionnaire data were collected at three Japanese sites with contrasting visibility and surrounding green coverage: KK Line in central Tokyo, JR Sakuragicho Station Square in Yokohama, and Tsukuba Center Square in Tsukuba. The results indicated two complementary pathways. First, visibility-related site contrasts were associated with more favorable perceptions, particularly prominence and brightness, which were in turn linked to more positive evaluations. Second, lower-green-coverage sites were associated with stronger stated preferences for introducing Mobile Green into everyday settings such as workplaces and homes. By contrast, preference for event-oriented deployment was relatively stronger in the civic plaza context of Tsukuba, suggesting that site meaning and routine use are also related to how temporary greening is interpreted. These findings indicate that the value of Mobile Green is context-dependent and suggest its potential as a flexible, temporary, and adaptive public-space intervention within broader strategies of sustainable urban transformation.

In the face of accelerating urbanization and climate change, the role of urban greening has expanded beyond esthetic and ecological functions to encompass contributions to human well-being, heat mitigation, and sustainable urban transformation[1–3]. The concept of green transformation (GX) has gained particular momentum as cities seek innovative and adaptive strategies for integrating green infrastructure into densely built environments[4,5]. Frantzeskaki[6] has argued that implementing such transformative agendas requires context-specific, flexible, and experimental approaches, often facilitated through urban living labs and nature-based solutions (NBS), in order to overcome institutional inertia and spatial constraints.

These insights underscore the importance of greening strategies that remain viable in constrained, impervious, and highly engineered urban spaces. Yet, the creation of new green space often faces substantial spatial and financial limitations, especially in cities experiencing intense development pressure[7,8]. As a result, attention has increasingly turned to underutilized urban open spaces such as rooftops, elevated highways, and plazas. Examples such as community-led rooftop greening in Singapore[9], the High Line in New York[10], and Seoul Skygarden[11] illustrate how former or unconventional urban surfaces can be reimagined as ecologically and culturally valuable public spaces.

At the same time, many such interventions are permanent and infrastructure-intensive, which may limit their scalability and adaptability across diverse urban settings. As Raymond et al.[12] and Kabisch et al.[13] emphasize, effective NBS should be multifunctional and adaptable. However, urban open spaces on artificial ground often lack the structural capacity required for conventional greening. Rooftops, viaducts, and decks constitute a substantial share of public space in dense cities but frequently remain underused because of regulatory, technical, and structural constraints[14,15]. These conditions call for lightweight and flexible greening approaches that can respond to diverse site conditions without major physical upgrades.

More broadly, such approaches can also be understood in relation to debates on temporary urbanism and tactical urbanism, which emphasize low-cost, non-permanent, and spatially adaptable interventions in public space[16,17]. In dense urban contexts where permanent transformation is difficult, temporary interventions can function as spatial experiments that test alternative spatial uses, reveal latent demand, and inform longer-term planning[17–19]. However, while these debates have expanded rapidly, limited

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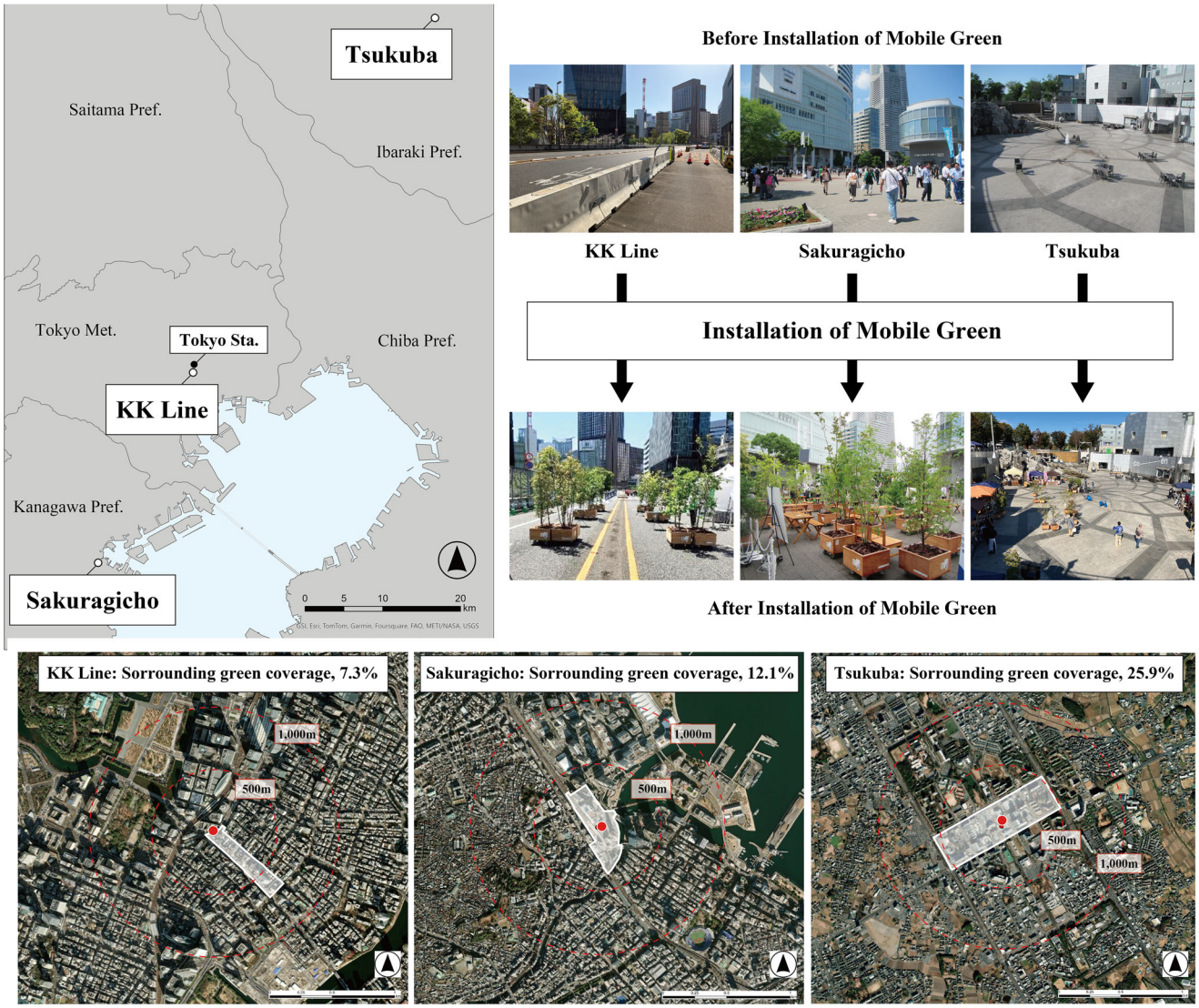


Fig. 1 | Study sites, Mobile Green installations, and contrasting surrounding green contexts. The study compared three Japanese urban sites where Mobile Green was temporarily installed: KK Line, an elevated expressway in central Tokyo; Sakuragicho Station Square, a station-front square in Yokohama; and Tsukuba Center Square, a civic plaza in Tsukuba. The upper panels show the regional locations of the three sites (left) and on-site views before and after installation (right). The lower panels present aerial images of the surrounding urban context, with the installation point marked by a red dot and dashed circles indicating 500 m and 1,000 m from the site center. The semi-transparent white polygons indicate the district areas used to calculate the surrounding green coverage ratios, which were 7.3%, 12.1%, and 25.9% for KK Line, Sakuragicho, and Tsukuba, respectively. The circular buffers are shown only to illustrate the spatial context around each site and do not represent the units used for the green coverage calculations. Basemap imagery in the lower panels: Esri, Vantor, Earthstar Geographics, and the GIS User Community.

attention has been paid to how temporary forms of greening are perceived and valued by users in different urban contexts.

In response to these challenges, this study focuses on "Mobile Green," a lightweight and movable greening system designed for flexible tree placement on artificial ground without infrastructure upgrades. Building on the modular green system developed by Sakamoto et al.[20], Mobile Green combines wheel-mounted carts and shallow-root planters to support mid-sized trees in reconfigurable arrangements. Although still under development and not commercially available, the system offers a potentially low-cost approach for introducing greenery into constrained urban spaces. Unlike conventional greenery, which is typically fixed and spatially integrated into a site, Mobile Green can be temporarily deployed, relocated, and rearranged according to site constraints and programmatic needs. This mobility makes it relevant not only as a greening device but also as a flexible public-space intervention.

This distinction raises an important question: how do people perceive and evaluate greenery when it appears in a temporary and movable form within highly engineered urban environments? Previous research has shown that green elements in public space are often associated with more favorable perceptions of comfort, attractiveness, and social interaction[21–23]. In particular, spaces in which greenery is more visually prominent tend to receive more positive evaluations from users[24]. At the same time, expressed preferences for greening may be stronger in neighborhoods with low ambient green coverage, where residents experience relative environmental deprivation[25]. Yet these relationships remain insufficiently examined in the case of temporary and movable greening interventions, whose meanings may depend not only on the green elements themselves but also on the contrast they create with the surrounding urban context[26,27].

To investigate these relationships, we compared three Japanese sites where Mobile Green was temporarily installed: KK Line, an elevated expressway in central Tokyo temporarily opened to pedestrians during a public access event; Sakuragicho, a station-front square in Yokohama; and Tsukuba Center Square, a civic plaza in Tsukuba (Fig. 1). These sites differ in the visibility of the installation, surrounding green coverage, and everyday

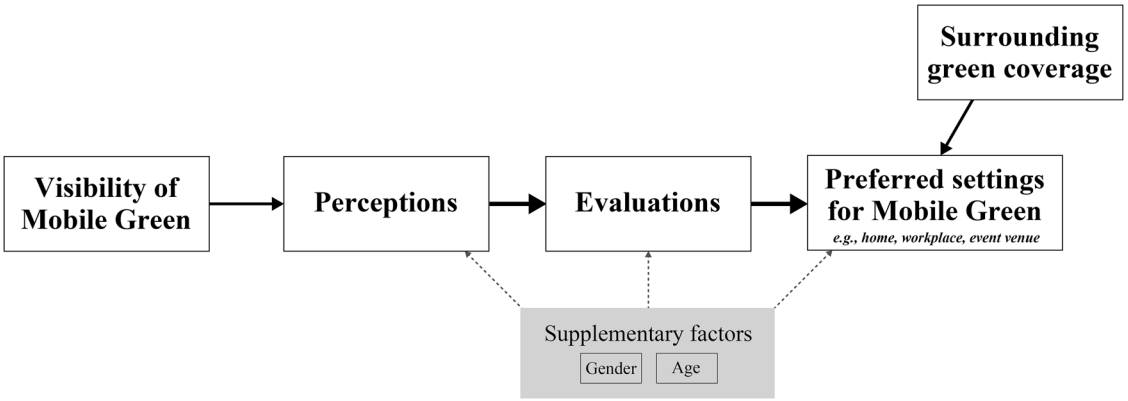


Fig. 2 | Conceptual framework of the study. Responses to Mobile Green are examined through two complementary pathways. First, the visibility of Mobile Green within a site is expected to be associated with perceptions, which in turn are associated with evaluations and preferred settings for introducing Mobile Green. Second, the surrounding level of green coverage is expected to be associated with preferred settings for Mobile Green. Gender and age are examined as supplementary factors that may also be associated with perceptions, evaluations, and preferred settings.

patterns of use, thereby providing contrasting urban contexts for examining perceptions, evaluations, and preferences related to Mobile Green.

Based on prior literature, we propose that responses to Mobile Green are shaped by two complementary pathways (Fig. 2). First, site-level differences in the visibility of the installation may be associated with more favorable perceptions, which may in turn contribute to more favorable evaluations. Second, site-level differences in the surrounding green context may be associated with stated preferences for where such greening should be introduced. In addition, gender and age are examined as supplementary factors that may also be associated with these responses. By analyzing user perceptions, evaluations, and preferences across these contrasting contexts, this study aims to clarify the conditions under which flexible greening interventions such as Mobile Green may be favorably received and desired, thereby contributing to evidence-based design and planning for sustainable urban transformation.

Results

Sample characteristics

A total of 312 responses were collected across the three study sites: 201 from KK Line, 53 from Sakuragicho, and 58 from Tsukuba. Gender distributions did not differ significantly across sites, whereas age distributions did, with KK Line including relatively more respondents in their 50 s and the other two sites relatively more respondents in younger age groups. Additional descriptive statistics for respondent characteristics are provided in Supplementary Tables 1 and 2.

Site differences in perceptions and evaluations

Perceptions and evaluations of Mobile Green-installed spaces varied across the three sites (see Supplementary Table 3). Based on post hoc comparisons, Sakuragicho scored significantly higher than Tsukuba on *prominent*, and significantly higher than both KK Line and Tsukuba on *bright*. In terms of evaluation, Sakuragicho also scored significantly higher than Tsukuba on *lively*. No similarly clear site-related contrasts were observed for the remaining perception items.

To examine which perception variables were most closely associated with the evaluation variables, correlations were calculated between the six perception items and the three evaluation items (see Supplementary Table 4). Among the six perception items, *prominent*, *bright*, and *relaxing* showed the clearest associations with positive evaluations and were therefore retained for the subsequent exploratory path analyses. In particular, *bright* was moderately to strongly associated with *exciting*, *lively*, and *like*, and was also positively correlated with both *prominent* and *relaxing*, indicating a central position within the observed correlation structure. By contrast, *prominent* showed a weaker direct association with *like*, although it was positively correlated with *bright*. On this basis, *prominent*, *bright*, and *relaxing* were retained as the perception variables in the subsequent models. The full set of questionnaire items and the distinction between retained and non-retained items are summarized in Supplementary Table 9.

Main observed pathways linking perceptions, evaluations, and preferred settings

Based on the preceding comparisons, exploratory path analyses within a structural equation modeling framework were used to examine the main observed associations among site-related conditions, selected perception variables, evaluations, and preferred settings for Mobile Green. For the visibility-related and evaluative pathway, the model focused on *prominent*, *bright*, and *relaxing*, which emerged as the most relevant perception variables in the preceding analyses.

To summarize the main observed relationships, Fig. 3 presents a simplified pathway structure, while detailed model estimates are reported in Tables 1 and 2. The exploratory path model for perceptions and evaluations showed the following descriptive fit indices: $\chi^2(10) = 9.60$, $P = 0.476$; CFI = 1.000; TLI = 1.002; RMSEA = 0.000; and SRMR = 0.027 (Table 1). Because the model specification was developed after preliminary inspection of the same dataset, these indices should be interpreted descriptively rather than as confirmatory evidence. Sakuragicho, representing the highest-visibility site, was positively associated with both *prominent* and *bright*, while KK Line showed a positive association with *prominent*. Female respondents were more likely to rate the installation as *relaxing*. Among the perception variables, *bright* and *relaxing* were positively associated with overall liking. *Prominent* was positively associated with *bright*, which was in turn positively associated with overall liking.

Before modeling preferred settings for Mobile Green, preliminary comparisons were conducted for both preferred installation settings and preferred vegetation types. No significant site differences were found in preferences regarding vegetation type. By contrast, significant differences emerged among the preferred installation settings (see Supplementary Table 5). In particular, *workplace* preference was higher at KK Line than at Tsukuba, *home* preference was higher at KK Line and Sakuragicho than at Tsukuba, and *event-venue* preference was relatively higher at Tsukuba than at the other two sites. Because these three settings showed the clearest differences across sites and age groups in the preliminary analyses (see Supplementary Table 7), the subsequent exploratory path analyses focused on *workplace*, *home*, and *event venue* as the preferred-setting outcomes.

The descriptive fit indices for the preferred-setting models are reported in Table 2. Across all three preferred settings, overall liking was positively associated with stated preferences for introducing Mobile Green. Site-specific associations were also observed. Preferences for *workplace* and *home* deployment were stronger at the lower-*green-coverage* sites, whereas preference for *event-venue* deployment was relatively higher at Tsukuba, the

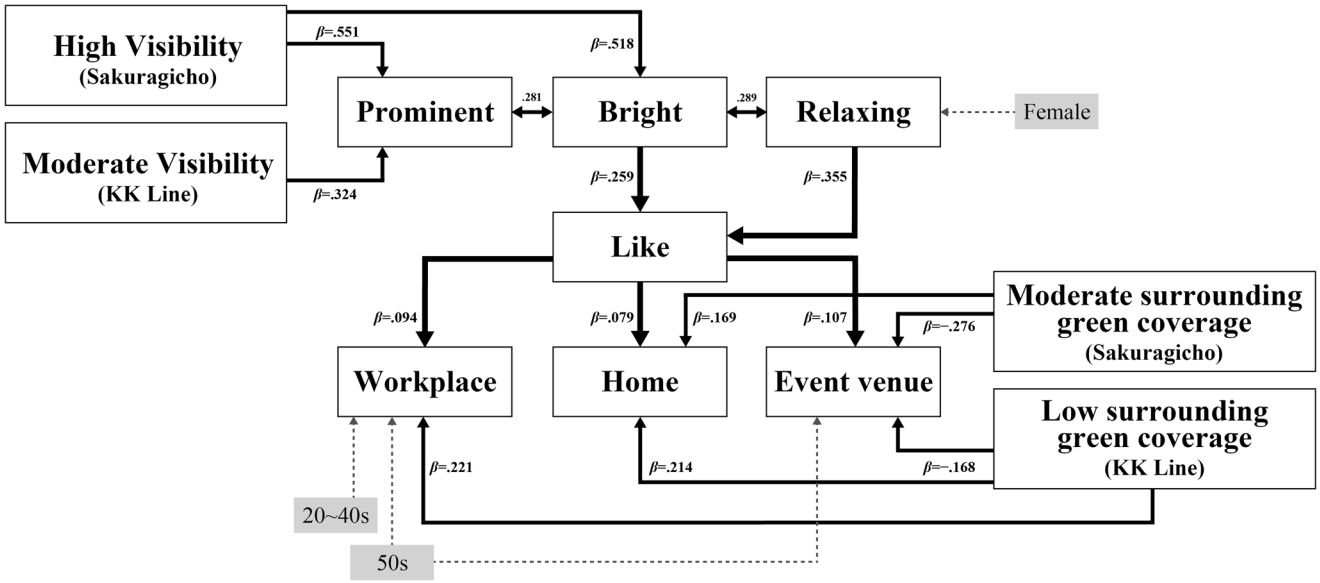


Fig. 3 | Simplified summary of the main associations identified in the exploratory path analyses linking site context, perceptions, and preferred settings for Mobile Green. Only the principal statistically significant associations are shown for clarity. Visibility-related site contrasts were associated with perceptions of Mobile Green: Sakuragicho, the highest-visibility site, was positively associated with both prominent and bright, whereas KK Line, the moderate-visibility site, was positively associated with prominent. Female respondents were more likely to perceive the installation as relaxing. Among the perception variables, bright and relaxing were positively associated with overall liking. Prominent was positively associated with bright, which was in turn positively associated with overall liking. Overall liking was positively associated with preferences for introducing Mobile Green in workplaces, homes, and event venues. Site-specific associations with preferred settings were interpreted in relation to differences in surrounding green coverage and place context: preferences for workplace and home deployment were stronger at the lower-green-coverage sites, whereas preference for event-venue deployment was relatively stronger at Tsukuba, the reference site with the highest surrounding green coverage. Age-group differences were also observed: respondents in their 20s–40 s showed stronger workplace preferences, and respondents in their 50 s showed stronger preferences for both workplace and event-venue settings. Tsukuba served as the reference category, representing the lowest visibility and highest surrounding green coverage among the three sites. Standardized path coefficients (β) are shown for the principal statistically significant regression paths, and standardized residual covariance coefficients are shown for the double-headed arrows among perception variables. The three preferred-setting outcomes were estimated in separate exploratory path models. These outcomes were selected from the full set of questionnaire options because they showed the clearest differences across sites and age groups in the preliminary analyses; the figure should therefore be interpreted as an exploratory summary rather than as a confirmatory model.

reference site with the highest *surrounding green coverage*. These site-specific associations were interpreted in relation to differences in *surrounding green coverage* and place context across the three sites. *Age-group* differences were also observed: respondents in their *20s–40 s* showed stronger *workplace* preferences, and respondents in their *50 s* showed stronger preferences for both *workplace* and *event-venue* settings.

Taken together, these results were broadly consistent with the conceptual framework shown in Fig. 2, while the exploratory nature of the models should be emphasized. Visibility-related site contrasts were associated primarily with perceptions of prominence and brightness. These perceptions were associated with overall liking, and overall liking was in turn associated with preferred settings for Mobile Green. At the same time, site-specific differences interpreted in relation to surrounding green coverage and place context were associated with preferred settings, especially for workplace and home deployment.

Discussion

The first set of observed associations in this study suggests that visibility-related site contrasts were associated with perceptions of Mobile Green, and that these perceptions were in turn linked to more favorable evaluations. In particular, the highest-visibility site, Sakuragicho, was positively associated with both prominence and brightness, while KK Line showed a weaker association with prominence. This pattern suggests that Mobile Green may be more likely to be noticed and positively interpreted when it is visually legible within a site. In this respect, the present findings are consistent with previous work showing that the spatial configuration and density of greenery can shape how people respond to urban green environments[20]. Related studies of plazas and parks have also shown that vegetation placement, especially along spatial edges or activity-supporting settings, can encourage lingering and enhance perceived attractiveness or comfort[28,29]. More broadly, these findings can also be read in light of landscape preference theories such as prospect-refuge and savanna theory[30,31], which suggest that people tend to respond positively to environments that combine legibility, interest, and a sense of environmental comfort.

A related finding concerns the relationship between perceptions and evaluation. Among the perception variables, brightness and relaxing impressions were positively associated with overall liking. Prominence showed a weaker association with liking in the model, although it was positively associated with brightness, which was itself positively associated with liking. This pattern suggests that visual salience alone may not be sufficient for favorable evaluation unless it is accompanied by qualities that are perceived as pleasant, bright or restorative. In addition, female respondents were more likely to perceive the installations as relaxing, which is broadly consistent with previous studies reporting gender-related differences in responses to green space[32–34]. These results also align with information-processing theory, which links perceptual coherence and preference[22], and with stress-reduction theory, which emphasizes the affective value of environments perceived as calming and restorative[21].

A second major set of observed associations links evaluation to preferred settings for Mobile Green. Across all three settings examined, overall liking was positively associated with stated preferences for introducing Mobile Green. This suggests that users who responded favorably to Mobile Green as an intervention were also more likely to imagine it being incorporated into a range of everyday or public settings. In this sense, overall evaluation may function as a general attitudinal bridge between on-site experience and broader demand for deployment.

At the same time, preferences were not associated with evaluation alone. Site-specific associations, interpreted in relation to surrounding green

Table 1 | Exploratory path-analysis results for site and gender differences in perceptions (prominent, bright, and relaxing) and the associations between perceptions and evaluations (exciting, lively, and like) of Mobile Green-installed spaces

Path	Estimate	Std.Error	*z*-value	*P*value	Std.Coef	Signif.
Site and gender differences in perceptions						
Prominent ~ KK Line	0.324	0.160	2.03	0.042	0.143	*
Prominent ~ Sakuragicho	0.551	0.203	2.71	0.007	0.191	**
Prominent ~ Female	−0.036	0.121	−0.299	0.765	−0.017	
Bright ~ KK Line	0.113	0.138	0.814	0.416	0.057	
Bright ~ Sakuragicho	0.518	0.176	2.94	0.003	0.205	**
Bright ~ Female	0.178	0.105	1.69	0.091	0.094	†
Relaxing ~ KK Line	−0.081	0.127	−0.632	0.528	−0.044	
Relaxing ~ Sakuragicho	−0.132	0.163	−0.813	0.416	−0.057	
Relaxing ~ Female	0.272	0.097	2.80	0.005	0.157	**
Associations between perceptions and evaluations						
Exciting ~ Prominent	0.191	0.040	4.77	<0.001	0.231	***
Exciting ~ Bright	0.275	0.049	5.60	<0.001	0.290	***
Exciting ~ Relaxing	0.334	0.051	6.51	<0.001	0.322	***
Lively ~ Prominent	0.216	0.047	4.59	<0.001	0.231	***
Lively ~ Bright	0.333	0.058	5.76	<0.001	0.310	***
Lively ~ Relaxing	0.263	0.060	4.36	<0.001	0.224	***
Like ~ Prominent	0.064	0.038	1.67	0.094	0.083	†
Like ~ Bright	0.259	0.047	5.54	<0.001	0.293	***
Like ~ Relaxing	0.355	0.049	7.27	<0.001	0.367	***
Residual covariances						
Prominent ~~ Bright	0.281	0.055	5.13	<0.001	0.282	***
Bright ~~ Relaxing	0.289	0.046	6.27	<0.001	0.364	***
Exciting ~~ Lively	0.172	0.036	4.71	<0.001	0.277	***
Exciting ~~ Like	0.135	0.029	4.59	<0.001	0.269	***
Lively ~~ Like	0.197	0.035	5.57	<0.001	0.332	***

Regression paths are reported using standard lavaan notation, with the dependent variable on the left-hand side of "~". Residual covariances are indicated by "~~". Tsukuba and male respondents are the reference categories. Unstandardized estimates and standardized coefficients are reported.
Significance codes: †$P < 0.10$, *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$.
Model fit: $\chi^2(10) = 9.60$, $P = 0.476$; CFI = 1.000, TLI = 1.002, RMSEA = 0.000, SRMR = 0.027.

coverage and place context, were also observed, especially for workplace and home settings. Preferences for these everyday settings were stronger at KK Line and, to a lesser extent, Sakuragicho than at Tsukuba, suggesting that Mobile Green may be particularly valued as a compensatory greening resource in environments where surrounding greenery is limited. This interpretation is consistent with previous studies showing that preferences for greenery, or willingness to pay for greener environments, tend to be stronger in urban areas where green exposure is relatively scarce[35–37]. The present findings extend this argument by suggesting that such compensatory demand may apply not only to permanent greenery, but also to flexible and temporary greening interventions such as Mobile Green.

A different pattern emerged for event-venue preference. In contrast to workplace and home settings, preference for installation in event settings was relatively stronger at Tsukuba than at KK Line and Sakuragicho. This suggests that preference for Mobile Green in event settings may not be driven primarily by green scarcity, but rather by the social meaning and routine use of the site. Tsukuba Center Square functions as a civic plaza that is already familiar as a setting for public events, and Mobile Green may therefore have been interpreted less as a compensatory everyday resource and more as a flexible design element for temporary activation. By contrast, KK Line is ordinarily inaccessible outside special events, and Sakuragicho functions primarily as a station-front square. In these settings, the idea of Mobile Green as an event-oriented intervention may have been less strongly anchored in existing patterns of public use.

Age-group differences were also selective rather than general. Preferences for workplace installation were stronger among respondents in their 20s–40 s, which plausibly reflects employment-related needs and the salience of everyday working environments. Preferences for event-venue installation were stronger among respondents in their 50 s, suggesting that interest in temporary greening for public or communal use may vary across life stages. Taken together, these findings suggest that demand for Mobile Green is shaped not only by how positively it is evaluated, but also by the interaction between surrounding green context, everyday life setting, and the social meaning of place. In this respect, Mobile Green may be useful not only where greenery is lacking, but also where temporary activation and programmatic flexibility are valued.

The present findings highlight the practical significance of Mobile Green not only as a greening device but also as a temporary, flexible, and adaptive public-space intervention. Across all three sites, Mobile Green could be delivered, installed, rearranged, and removed within highly engineered urban settings, including an elevated expressway, a station-front square, and a civic plaza. Despite its non-permanent character, the installations were positively received, suggesting that temporary greening can be associated with meaningful perceptual and evaluative responses even in spaces where conventional planted greenery would be difficult to introduce. In this respect, Mobile Green appears to offer a mode of intervention that is distinct from fixed and site-integrated greening: its value lies not only in providing vegetation, but also in

Table 2 | Exploratory path-analysis results for site differences, age-group differences, and associations between overall liking and preferences for introducing Mobile Green in workplace, home, and event-venue settings

Path	Estimate	Std.Error	z-value	*P*value	Std.Coef	Signif.
Preference for workplace deployment						
Workplace ~ Like	0.094	0.031	3.01	0.003	0.161	**
Workplace ~ KK Line	0.221	0.071	3.11	0.002	0.217	**
Workplace ~ Sakuragicho	0.129	0.088	1.46	0.144	0.099	
Workplace ~ 20 s	0.305	0.124	2.46	0.014	0.188	*
Workplace ~ 30 s	0.313	0.111	2.82	0.005	0.249	**
Workplace ~ 40 s	0.318	0.111	2.87	0.004	0.253	**
Workplace ~ 50 s	0.331	0.105	3.15	0.002	0.309	**
Workplace ~ 60 s	0.122	0.113	1.07	0.284	0.092	
Preference for home deployment						
Home ~ Like	0.079	0.03	2.66	0.008	0.144	**
Home ~ KK Line	0.214	0.067	3.18	0.001	0.224	***
Home ~ Sakuragicho	0.169	0.084	2.02	0.044	0.139	*
Home ~ 20 s	−0.033	0.118	−0.281	0.779	−0.022	
Home ~ 30 s	−0.172	0.105	−1.63	0.102	−0.145	
Home ~ 40 s	−0.026	0.105	−0.245	0.806	−0.022	
Home ~ 50 s	−0.118	0.099	−1.19	0.234	−0.118	
Home ~ 60 s	0.069	0.108	0.641	0.522	0.055	
Preference for event-venue deployment						
Event venue ~ Like	0.107	0.030	3.53	<0.001	0.189	***
Event venue ~ KK Line	−0.168	0.069	−2.45	0.014	−0.171	*
Event venue ~ Sakuragicho	−0.276	0.086	−3.22	0.001	−0.219	**
Event venue ~ 20 s	0.177	0.121	1.47	0.142	0.112	
Event venue ~ 30 s	0.208	0.108	1.93	0.054	0.170	†
Event venue ~ 40 s	0.178	0.107	1.65	0.098	0.145	†
Event venue ~ 50 s	0.256	0.102	2.52	0.012	0.247	*
Event venue ~ 60 s	0.019	0.110	0.169	0.866	0.014	

Regression paths are reported using standard lavaan notation, with the dependent variable on the left-hand side of "~". Tsukuba is the reference category for site. Respondents in their teens, 70 s, and 80 s or older constitute the combined reference group for age. Unstandardized estimates and standardized coefficients are reported. Significance codes: †P < 0.10, *P < 0.05, **P < 0.01, ***P < 0.001. Model fit for each exploratory path model: Workplace model: $\chi^2(28) = 24.76$, P = 0.641; CFI = 1.000, TLI = 1.024, RMSEA = 0.000, SRMR = 0.031. Home model: $\chi^2(28) = 38.22$, $P = 0.094$; CFI = 0.959, TLI = 0.927, RMSEA = 0.034, SRMR = 0.038. Event-venue model: $\chi^2(28) = 24.47$, $P = 0.657$; CFI = 1.000, TLI = 1.026, RMSEA = 0.000, SRMR = 0.030.
The three setting-specific outcomes were estimated in separate models. See also Table 1 for estimates of perceptions and evaluations.

temporarily transforming how urban space is perceived, used, and imagined.

These implications are particularly relevant in relation to urban sustainability and the broader GX agenda. In dense urban environments where land is scarce, contested, or structurally constrained, conventional green infrastructure is often difficult to implement. Because Mobile Green is movable, reconfigurable, and deployable without major infrastructure modification, it offers a practical complement to conventional greening in such contexts. This flexibility aligns closely with key principles of nature-based solutions (NBS) and green infrastructure (GI), especially multifunctionality, adaptability, and socioecological resilience[6,13,38]. Rather than replacing conventional greening, Mobile Green may therefore function as a complementary strategy that expands the range of urban spaces in which greening can be experimentally introduced and publicly experienced.

This point is especially important for underused, transitional, or liminal urban spaces. Recent discussions in NBS and adaptive urban design have emphasized the value of bringing ecological functions into temporary or otherwise difficult-to-green settings such as rooftops, decks, and transitional plazas[39]. The present study suggests that Mobile Green speaks to this agenda. Because it can be installed and removed without permanent spatial reallocation, it may be used not only to provide temporary environmental improvement but also to test spatial scenarios, support short-term activation, and identify the urban contexts in which users most strongly value greening. In this sense, Mobile Green may serve as a practical bridge between ecological intervention and experimental public-space design.

The planning value of Mobile Green nevertheless appears to be context-dependent. In green-scarce and highly built settings, it may function most strongly as a compensatory greening resource for everyday environments such as workplaces and homes. In highly visible public spaces, its role may lie in its immediate perceptual salience. In civic or event-oriented spaces, it may operate as a flexible design element that supports temporary activation and place-making. This implies that temporary greening should not be understood as a one-size-fits-all solution. Rather, interventions such as Mobile Green may be most useful when their spatial form, placement, and intended use are matched to the surrounding green context and to the social meaning of the site.

In addition, the observed differences by gender and age point to broader questions of inclusivity and equity in green-space planning[40,41]. Although these demographic associations were supplementary rather than dominant, they suggest that preferences for flexible greening are not socially uniform. The stronger relaxing responses among female respondents, and the selective age-related preferences for workplace and event settings, indicate that temporary greening interventions may be experienced differently depending on lifestyle context, everyday routine, and stage of life. This

implies that adaptive GI technologies should be planned not only in relation to physical site conditions, but also in relation to the diverse values and needs of urban users.

Several limitations should nevertheless be acknowledged. First, the study relied on a cross-sectional survey design and on self-reported perceptions, evaluations, and preferences. The observed relationships should therefore be interpreted as associative rather than causal. Second, the path models were exploratory. The variables included in the main models were selected after preliminary inspection of descriptive statistics, site-level differences, and correlation patterns in the same dataset used for model estimation. This procedure may inflate model-fit indices and the apparent statistical strength of some associations. The reported pathways should therefore be interpreted as hypothesis-generating rather than confirmatory. Future studies should examine the robustness of these associations using independently collected samples, pre-specified model structures, and a wider range of urban contexts.

Third, visibility was treated comparatively across the three sites rather than measured directly using techniques such as viewshed analysis or image-based greenness indicators. Fourth, while surrounding green coverage was quantified at the district level, differences among the three sites were not limited to this factor alone. Spatial layout, deployment density, accessibility, surrounding infrastructure, temporal use patterns, and broader place-based meanings may also have shaped responses. Accordingly, site differences should not be attributed solely to visibility and surrounding green coverage.

These limitations are compounded by several additional methodological considerations. Sample sizes differed substantially across the three sites, which may have affected the precision of site-level comparisons. Moreover, because the survey was conducted during public events and relied on voluntary participation, respondents who were more interested in greenery, public events, or the intervention itself may have been overrepresented. The installations were also not fully standardized across sites in terms of density, arrangement, and event context, which may have introduced further confounding variation. These limitations do not negate the observed patterns, but they do suggest that the findings should be interpreted with caution and as an initial empirical step rather than as definitive evidence of causal effects.

Future research should therefore move beyond cross-sectional self-report measures alone. More robust designs could include pre–post comparisons, repeated observations across seasons, or controlled comparisons in which visibility is manipulated within the same site or surrounding green context is varied across otherwise comparable settings. It would also be valuable to incorporate behavioral indicators, such as time spent, movement patterns, or repeat visitation, as well as physiological measures related to affective or restorative responses. Such approaches would make it possible to evaluate more precisely how temporary greening interventions such as Mobile Green are associated with both immediate perception and longer-term patterns of public-space use.

Methods

Study sites and installation contexts

This study was conducted at three urban sites in Japan where the "Mobile Green" system was temporarily installed: KK Line in central Tokyo, JR Sakuragicho Station Square in Yokohama, and Tsukuba Center Square in Tsukuba. As shown in Fig. 1, the three sites differ in surrounding green coverage, visibility of the installation, and everyday patterns of public use, thereby providing contrasting contexts for examining perceptions, evaluations, and preferred settings for Mobile Green.

KK Line is an elevated expressway in central Tokyo that is ordinarily closed to pedestrians. Mobile Green was installed during the 2024 GINZA SKY WALK event (4–6 May 2024), when the expressway was temporarily opened for public access. Although a large green area associated with the Imperial Palace lies to the north, the area immediately surrounding the site is densely built and contains very limited publicly accessible greenery. This site therefore represents a highly engineered urban setting with the lowest surrounding green coverage among the three sites.

JR Sakuragicho Station Square is a station-front public space in Yokohama that functions as a gateway to the surrounding business and tourism district. Mobile Green was installed there during a public event related to GREEN×EXPO 2027 on 22–23 June 2024. Like KK Line, the site is embedded in a highly urbanized setting, but compared with KK Line it is associated with somewhat greater surrounding greenery, including major streets lined with large-canopy street trees. Among the three sites, Sakuragicho was also the location where the installation was most visually prominent.

Tsukuba Center Square is a civic plaza in the center of Tsukuba that is used for everyday public activity as well as cultural and community events. Mobile Green was installed during a weekend coffee festival on 9–10 November 2024, shortly after the square had been renewed as part of a broader urban revitalization effort. In contrast to the other two sites, areas within a short distance of the square include a mixture of residential land, open land, and agricultural land, resulting in greater surrounding green coverage. This site therefore represents the greenest context among the three locations and was also the one most strongly associated with routine event use.

Mobile Green intervention and site-specific conditions

Mobile Green is a modular greening system designed to enable the temporary and flexible placement of small trees in urban public spaces. The system combines a lightweight rooftop-greening tray with a mobile wooden cart, allowing trees to be transported, arranged, and removed without permanent ground intervention.

Each tree was planted in a 53 × 53 cm MEP® tray (Elmich, Singapore) with a soil depth of 15 cm. The planted trees were approximately 2 m in height and were cultivated with their roots fixed to a specially prepared fiber base to ensure stability during pre-cultivation, transport, and temporary installation. The tray system also included an internal water reservoir with a capacity of up to 7 L, reducing the need for frequent irrigation.

The trays were mounted on 56 × 56 cm wooden carts with a height of 45 cm. Each cart was equipped with a brake mechanism, allowing safe relocation as well as stable installation in a range of public-space settings.

In total, 18 tree species were used across the Mobile Green installations, including 8 evergreen and 10 deciduous species. Detailed information on species composition and system configuration is provided in Table 3.

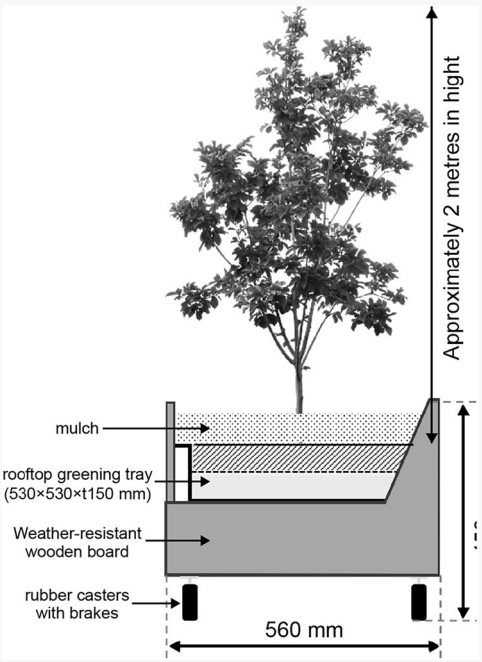
The total deployable area for Mobile Green was 60 m^2 at KK Line, 69 m^2 at Sakuragicho, and 1287 m^2 at Tsukuba. The number of trees installed at each site was 24, 30, and 30, respectively, corresponding to tree densities of 0.40 trees/m^2 at KK Line, 0.43 trees/m^2 at Sakuragicho, and 0.019 trees/m^2 at Tsukuba. Because the available space at Tsukuba was substantially larger than at the other two sites, the same number of trees resulted in a much lower planting density and lower visual salience within the site.

Although KK Line and Sakuragicho had similar tree densities, their spatial arrangements differed substantially. At Sakuragicho, Mobile Green units were distributed more evenly across the plaza, allowing the installation to be seen from multiple positions within the site. At KK Line, by contrast, trees were arranged in island-like clusters along the edges in order to preserve pedestrian flow on the narrow elevated roadway, which reduced within-site visibility relative to Sakuragicho. Based on planting density and spatial arrangement, the visibility of Mobile Green was treated comparatively as highest at Sakuragicho, moderate at KK Line, and lowest at Tsukuba.

In contrast, surrounding green coverage was highest at Tsukuba and lowest at KK Line. According to Kiyono et al.[42]., the green coverage ratios of the corresponding districts were 25.9% for Azuma 1-chome (Tsukuba), 12.1% for Sakuragicho 1-chome (Yokohama), and 7.3% for Ginza 4-chome (Tokyo), where KK Line is located. These values were calculated at the district level and were used to characterize the surrounding green context of each site. The geographic locations of the three study sites, along with aerial images, district polygons used for the green coverage calculations, and photographs taken before and after the installation of Mobile Green, are

Table 3 | Tree species composition and system configuration of the Mobile Green installations

Species	KK Line	Sakuragicho	Tsukuba	System configuration of Mobile Green
Evergreen				
Fraxinus griffithii	3	4	6	
Quercus myrsinifolia	3	3	6	
Myrica rubra	1	2	3	
Cornus florida var. urbiniana	2	1	2	
Cornus kousa	0	0	1	
Abies firma	0	0	1	
Juniperus scopulorum 'Blue Heaven'	0	0	1	
Chamaecyparis pisifera	0	0	1	
Deciduous				
Lagerstroemia indica	2	3	4	
Fraxinus lanuginosa	2	5	0	
Pseudocydonia sinensis	2	2	2	
Cercidiphyllum japonicum	2	3	0	
Stewartia pseudocamellia	2	3	0	
Acer palmatum	1	1	1	
Magnolia stellata	1	1	1	
Eucommia ulmoides	1	1	1	
Cornus florida	1	1	0	
Stewartia monadelpha	1	0	0	
Total	24	30	30	



shown in Fig. 1. The species composition and number of trees installed at each site are summarized in Table 3.

Questionnaire survey and measures

A questionnaire survey was conducted to examine users' perceptions, evaluations, and preferences regarding Mobile Green, including preferred future deployment settings and vegetation types. To assess perceptions and evaluations of the installations, the questionnaire drew on adjective pairs used in Sakamoto et al.[20]., which had previously been applied to modular green interventions. Six adjective pairs were used to assess spatial perceptions (*subtle–prominent*, *narrow–spacious*, *enclosed–open*, *dark–bright*, *stifling–refreshing*, *tense–relaxing*), and three adjective pairs were used to assess overall evaluations of the space (*boring–exciting*, *desolate–lively*, *dislike–like*). All items were rated on a six-point Likert scale.

To examine preferred settings for future deployment, respondents were also asked where Mobile Green should be introduced, with multiple selections permitted among five location types: *event venues*, *open cafés*, *public parks*, *workplaces*, and *homes*. Preferences regarding vegetation type were assessed through seven selectable options: *more plants in general*, *flowering trees*, *fruit-bearing trees*, *lawns or flowering groundcover*, *larger trees*, *smaller trees*, and *no plants needed*. Demographic information was also collected, including gender (*female*, *male*, *prefer not to say*) and age group (10-year intervals from *teens* to *80 years and above*).

The survey was administered during each installation event using stickers printed with a QR code that linked to an online questionnaire form. Stickers were distributed to site users at KK Line (4–6 May 2024), Sakuragicho (22–23 June 2024), and Tsukuba (9–10 November 2024). Participation was voluntary, and respondents completed the survey on their own mobile devices during or shortly after the event. A total of 312 responses were collected: 201 from KK Line, 53 from Sakuragicho, and 58 from Tsukuba.

The questionnaire survey was conducted independently by Sumitomo Forestry Co., Ltd. to collect reference information for the development and improvement of the Mobile Green system and in accordance with the company's internal procedures for anonymous questionnaires. The authors subsequently used the anonymized questionnaire data for research purposes; the survey did not collect personally identifying information. Participation was voluntary, and potential respondents were informed orally and on the introductory page of the online questionnaire that they should proceed only if they agreed to participate. Completion and submission of the questionnaire were treated as consent to participate. The Environment, Health and Safety Office at the University of Tokyo confirmed that the present study did not fall within the scope of formal ethical review because it involved the secondary research use of existing anonymized data originally collected by Sumitomo Forestry Co., Ltd. for operational purposes. No reference number was issued for this determination.

Structural equation modeling

Exploratory path analyses within a structural equation modeling (SEM) framework were used to examine associations among site-related conditions, individual attributes, perceptions, evaluations, and preferred settings for Mobile Green, following the conceptual framework shown in Fig. 2. The analyses were conducted in R using the lavaan package (version 0.6-21) and were estimated using maximum likelihood (ML). All variables were treated as observed variables.

The analysis was conducted in two stages. In the first-stage model, perception variables were modeled as predictors of evaluation variables. Although six perception items were measured in the questionnaire, the model focused on *prominent*, *bright*, and *relaxing*, which showed the clearest site-related contrasts and associations with the evaluation variables in the preliminary descriptive and correlational analyses (see Supplementary Table 4). Evaluation variables were *exciting*, *lively*, and *like*. *Site-related visibility* contrasts were represented by dummy variables for KK Line and Sakuragicho, with Tsukuba as the reference category, and *gender* was included as an additional explanatory variable. Residual covariances among selected perception variables and among evaluation variables were estimated explicitly.

In the second-stage models, preferences for future deployment settings were examined. Although five setting options were included in the questionnaire, the models focused on *workplace*, *home*, and *event venue*, as these

settings showed the clearest differences across sites and age groups in the preliminary comparative analyses (see Supplementary Table 5). To maintain a parsimonious pathway structure, *like* was treated as a representative evaluation variable linking perceptions and preferred settings. Each preferred-setting outcome was estimated in a separate exploratory path model. In each model, overall liking, site contrasts, and age-group contrasts were included as explanatory variables. The site contrasts were interpreted in relation to differences in surrounding green coverage and place context across the three sites; surrounding green coverage was not entered into the models as a continuous variable.

The main path models were developed after preliminary inspection of descriptive statistics, site-level differences, and correlation patterns in the same dataset. Accordingly, the reported fit indices and P values should be interpreted descriptively rather than as confirmatory evidence. The full set of questionnaire items and the distinction between retained and non-retained items are reported in Supplementary Table 9, and the lavaan model specifications are provided in Supplementary Note 3.

There were no missing values in the analyzed questionnaire items because all items were set as required in the online survey form. Model fit was described using the chi-square statistic, comparative fit index (CFI), Tucker–Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR). Detailed path coefficients and model estimates are reported in Tables 1 and 2, with additional model details provided in the Supplementary Information.

Mandated checklist

No mandated reporting checklist applies to this study design.

Data availability

The data that support the findings of this study consist of questionnaire responses collected from human participants. Owing to ethical restrictions related to participant privacy and consent, the raw individual-level data are not publicly available. Aggregated data supporting the findings of this study, including respondent characteristics by site, site-level descriptive statistics for perception and evaluation items, correlations among perception and evaluation items, and summaries of preferences by site and respondent characteristics, are provided in the Supplementary Information. Additional aggregated data may be made available by the corresponding author upon reasonable request, where this is consistent with the consent provided by participants.

Code availability

No custom software was developed for this study. All data processing and statistical analyses were conducted in R, including exploratory path analyses using the lavaan package (version 0.6-21). The R scripts used to generate the reported descriptive statistics and exploratory path-analysis results, including the lavaan model specifications, are available from the corresponding author upon reasonable request. The lavaan model specifications are also provided in the Supplementary Information.

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Author contributions

K.S. conceived and designed the overall study, secured research funding, analyzed the data, and wrote the manuscript. S.I., H.I., and K.N. contributed to the preparation of the test sites and materials for the field experiments. M.Y. contributed to the conception of the study and the acquisition of research funding.

Competing interests

Some authors are affiliated with Sumitomo Forestry Co., Ltd., which is involved in the research and development and potential future commercialization of the Mobile Green system examined in this study. This study was conducted as a collaborative research project with Sumitomo Forestry Co., Ltd. and received funding from the company under a collaborative research agreement. Sumitomo Forestry Co., Ltd. also filed a patent application related to Mobile Green on 29 January 2024 (application no. 2024-010683; currently pending and not yet published). H.I., S.I., and K.N. are listed as inventors on this application. The patent relates to the internal structure of the wheeled cart used in the Mobile Green system. Among these authors, K.N. is responsible for the commercialization project, and S.I. and H.I. are involved in system development. The authors have no employment, consultancy, commissioned research, advisory, or joint business relationships with the external organizations involved in the demonstration experiments, including the Tokyo Metropolitan Government, Tokyo Expressway Co., Ltd., the Japan Association for the 2027 International Horticultural Expo, and Tsukuba City, beyond the conduct of this research. All other authors declare no competing financial or non-financial interests.

Additional information

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